
Query-Efficient Quantum Approximate Optimization via Graph-Conditioned Trust Regions

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Abstract

Classical optimization can dominate the measurement cost of variational quantum algorithms, yet optimizer routing may be less reliable than a fixed schedule. We formulate graph-conditioned trust regions as a target-free, fixed-budget portfolio for depth-two MaxCut QAOA. Nine arms receive identical 32-call ceilings. A ridge model predicts regret, a family control supplies a fallback, and a residual order statistic abstains when evidence is insufficient. After a 64-graph development audit, the selector is trained on 48 graphs, calibrated on 48 disjoint graphs, and locked before evaluation on 160 instances. Gated deployment yields mean regret 0.16930 versus 0.17207 for the fallback, but accepts only 12 routes, four harmful. One-sided coverage is 0.88125, below the frozen 0.90 criterion; the reliability claim therefore fails. Fixed family quotas also preclude pooled exchangeability. The result is a reproducible exact-statevector failure analysis, not evidence of hardware, scaling, or quantum advantage.

1 Introduction

Variational quantum algorithms combine a parameterized circuit with a classical optimizer [41, 46]. Their near-term appeal comes with a basic cost problem: every black-box objective query can require many circuit preparations and measurements [10, 48]. The optimizer is therefore part of the physical resource model rather than an interchangeable software detail [14].

The quantum approximate optimization algorithm (QAOA) makes the interaction explicit by alternating cost and mixer unitaries before optimizing a small angle vector [22, 23]. Its empirical behavior depends on graph structure, depth, initialization, and search budget [65]. Parameter concentration can make fixed schedules competitive on typical instances [2, 12]; warm starts can inject classical solutions [20]; and transferred parameters can exploit similarity across weighted graphs [53].

Graph-conditioned optimization attempts to use this shared structure before or during search. A predictor can propose angles, a local covariance, a trust region, or an optimizer choice. Graph neural initialization demonstrates the broader opportunity [31], and learned variational optimizers show that trajectories themselves may be predicted [33]. The statistical question is less often

made explicit: when should a learned recommendation replace a strong, cheap control?

That question cannot be answered by average terminal quality alone. Hyperparameter and model selection on the same benchmark used for evaluation can introduce substantial optimism [13]. A router can improve a pooled average by helping one family while harming another, and a target-based stopping rule can depend on an oracle unavailable in deployment. The classical algorithm-selection problem already separates instance features, candidate algorithms, and performance measures [49]; the quantum setting adds an unusually costly response surface.

We therefore replace privileged target hitting with a fixed-query anytime metric. Every arm receives 32 objective calls, and utility is the average normalized regret over checkpoints $\{1, 2, 4, 8, 16, 32\}$. A relabeling-invariant structural model predicts each arm’s utility. The strongest development arm within the known graph family remains the fallback. Routing is accepted only when an upper residual order statistic indicates that the predicted selected arm should beat that fallback.

This abstention mechanism is inspired by split-conformal reasoning [3, 59]. Its finite-sample interpretation requires exchangeability, however, and the released confirmatory design uses fixed family-size quotas with sequential non-isomorphism filtering. We therefore report both the frozen rule and the limits of its coverage interpretation [8].

Figure 1 shows the evidence chronology. Development fits the selector and family fallback; calibration fixes one residual order statistic; the graph list, code, configuration, and decision are hashed; and the audit is evaluated once. This chronology is as important as the model because unrecorded iteration on audit outcomes would erase the meaning of a confirmatory result.

The final decision is negative. The deployed policy slightly improves mean anytime regret, but it accepts only 12 of 160 routes, harms four accepted instances, and misses the frozen empirical-coverage threshold. A result can have a favorable mean and still fail its reliability criterion. Preserving that distinction is the paper’s central contribution.

An internally locked, target-free decision path

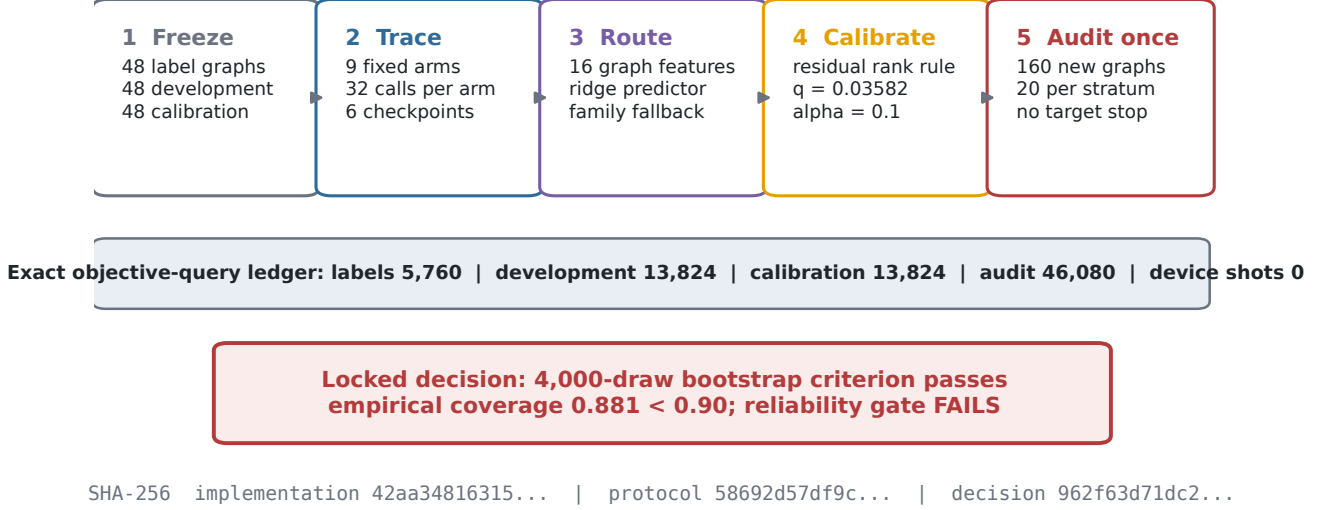


Figure 1: **Internally locked fixed-budget protocol.** Development data fit the structural selector and family fallback; calibration fixes a one-sided residual order statistic. The graph list, implementation, configuration, and decision are hashed before the 160-graph audit is evaluated.

2 Background

2.1 Quantum Approximate Optimization

For an unweighted graph $G = (V, E)$, define

$$C_G = \sum_{(u,v) \in E} \frac{I - Z_u Z_v}{2}, \quad B = \sum_{v \in V} X_v. \quad (1)$$

A depth- p QAOA state is

$$|\gamma, \beta\rangle = \prod_{j=p}^1 e^{-i\beta_j B} e^{-i\gamma_j C_G} |+\rangle^{\otimes |V|}. \quad (2)$$

The objective is $F_G(\theta) = \langle \gamma, \beta | C_G | \gamma, \beta \rangle$. Analytic parameter-shift rules can evaluate gradients on compatible hardware [51, 61], while stochastic estimators trade coordinate resolution for fewer objective calls [55].

Gradient methods do not eliminate trainability concerns. Random or highly expressive circuits can exhibit barren plateaus [42]; expressibility is connected to gradient concentration [28]; and control-theoretic tools refine the diagnosis [35]. Gradient-free search also degrades in flat landscapes [4]. Our study therefore compares complete fixed-budget trajectories rather than assuming one access model is uniformly best.

2.2 Fixed-Budget Black-Box Optimization

At checkpoints

$$\mathcal{T} = \{1, 2, 4, 8, 16, 32\}, \quad (3)$$

arm a has post hoc normalized regret

$$r_{ia}(t) = 1 - \frac{\max_{s \leq t} F_{G_i}(\theta_{ias})}{C_{\max}(G_i)} \quad (4)$$

and utility

$$u_{ia} = \text{AURC}_{ia} = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} r_{ia}(t). \quad (5)$$

Lower is better. Exact MaxCut $C_{\max}$ is used only after optimization; no arm sees a target or target-hitting event.

The portfolio spans fixed schedules, transferred angles, multistart, SPSA, and the graph-conditioned trust-region arm. This heterogeneity reflects the fact that optimizer rankings depend on the variational problem and budget [11]. Mesh-adaptive search provides a principled derivative-free alternative [5]; BOBYQA uses bounded quadratic interpolation [47]; and covariance-matrix adaptation learns a search distribution [26]. We freeze simpler arms because a portfolio is only meaningful relative to its actual controls.

2.3 Selective Routing and Risk

Let $x(G) \in \mathbb{R}^{16}$ be a relabeling-invariant graph summary and $\hat{u}_a(x)$ the predicted utility of arm a . The proposed

arm is

$$s(x) = \arg \min_a \hat{u}_a(x). \quad (6)$$

For family label f , the fallback $b(f)$ is the development arm with minimum family-mean utility. Define the realized selected-minus-fallback regret

$$\Delta = u_s - u_b \quad (7)$$

and prediction residual

$$R = \Delta - (\hat{u}_s - \hat{u}_b). \quad (8)$$

Given a calibration upper order statistic q , routing is accepted when

$$\hat{u}_s - \hat{u}_b + q < 0. \quad (9)$$

Otherwise the system abstains to $b(f)$.

This is selective prediction: coverage, acceptance, joint harm, and conditional harm are different quantities. Conformalized quantile regression can adapt interval width to covariates [50]; online selective methods can target error criteria under other assumptions; and classical exact binomial limits can summarize observed event rates [15]. None licenses a coverage claim without a sampling model.

3 Related Work

QAOA parameter initialization exploits concentration, interpolation, and problem structure [12, 65]. Learned graph initializers extend this idea using message passing [31]. Weighted-MaxCut parameter transfer addresses related objectives [53], while ramp schedules provide a strong low-complexity comparator [43]. Our portfolio treats these ideas as arms or controls rather than assuming one initialization dominates.

Graph convolutional networks use normalized neighborhood aggregation [34]; graph attention learns neighbor weights [58]; and graph-isomorphism networks characterize a powerful message-passing class [63]. We intentionally use 16 hand-computed invariant summaries with ridge regression. This low-capacity choice reduces model flexibility and makes every routing score inspectable.

Bayesian optimization is another approach to expensive black-box search. Efficient global optimization introduced improvement-based acquisition [32]; practical machine-learning formulations popularized Gaussian-process tuning [54]; and a broad review organizes surrogate and acquisition choices [52]. Differentiable Monte Carlo acquisitions are available in BoTorch [6]. These methods may be useful arms, but fitting a surrogate does not make its objective evaluations free.

Algorithm configuration learns settings across tasks. Sequential model-based configuration uses prior runs to propose new configurations [30], and SMAC3 exposes a reproducible modern package [38]. The portfolio view also connects to active subspaces [16], sliced inverse regression [36], and regression graphics [17], but our predictor

reduces neither the circuit parameter dimension nor the graph feature dimension adaptively.

Randomized linear algebra is relevant to possible future low-rank structural models. Randomized range finding provides scalable approximation [25]; fixed-precision blocked QB supports adaptive rank [64]; matrix sketching offers streaming alternatives [37, 62]; and matrix concentration supplies probabilistic tools under explicit assumptions [57]. The released ridge model is small enough that none of these approximations is needed.

Finally, statistical safeguards address distinct risks. Bootstrap resampling estimates sampling variation [19]; multiple-testing procedures control false discoveries under stated dependence conditions [9]; and the Neyman–Pearson framework separates size from power [44]. Multiple-data-set comparisons also require analysis at the data-set level [18]. Our locked gate is a decision protocol, not a post hoc collection of significance tests.

4 Shortcomings of Unconditional Structural Routing

4.1 Selection Bias and Distribution Shift

Choosing a route because it is best on development data is not a distribution-free argument. Even a perfect empirical ranking can reverse under task shift, a basic distinction in algorithm selection and honest model assessment [13, 49].

Theorem 4.1 (Near-maximal bounded excess loss after routing shift). *For every $\varepsilon \in (0, 1)$, there exist two optimizer arms a, b , losses in $[0, 1]$, a development distribution P_0 , and a deployment distribution P_1 such that a is the unique risk minimizer under P_0 , b is the unique risk minimizer under P_1 , and routing unconditionally to a has deployment excess risk $1 - \varepsilon$.*

Appendix A.1 gives a complete construction. The theorem is elementary by design: no model complexity is required for a failure. This is why graph splits, family strata, and chronology must be stated before interpreting a learned router.

The first version of the project optimized against an offline target and compared a learned trust-region policy primarily with random restart. Stronger fixed schedules were cheaper, and the target was unavailable at deployment. The current portfolio keeps the original evidence as a legacy diagnostic but does not combine it with the target-free confirmatory result.

4.2 Coverage and Conditional Harm

Let A be the event that the selector accepts a route and $H = \{\Delta > 0\}$ the event that the selected arm is worse than the fallback.

One-shot confirmatory audit: a small average gain without reliable routing

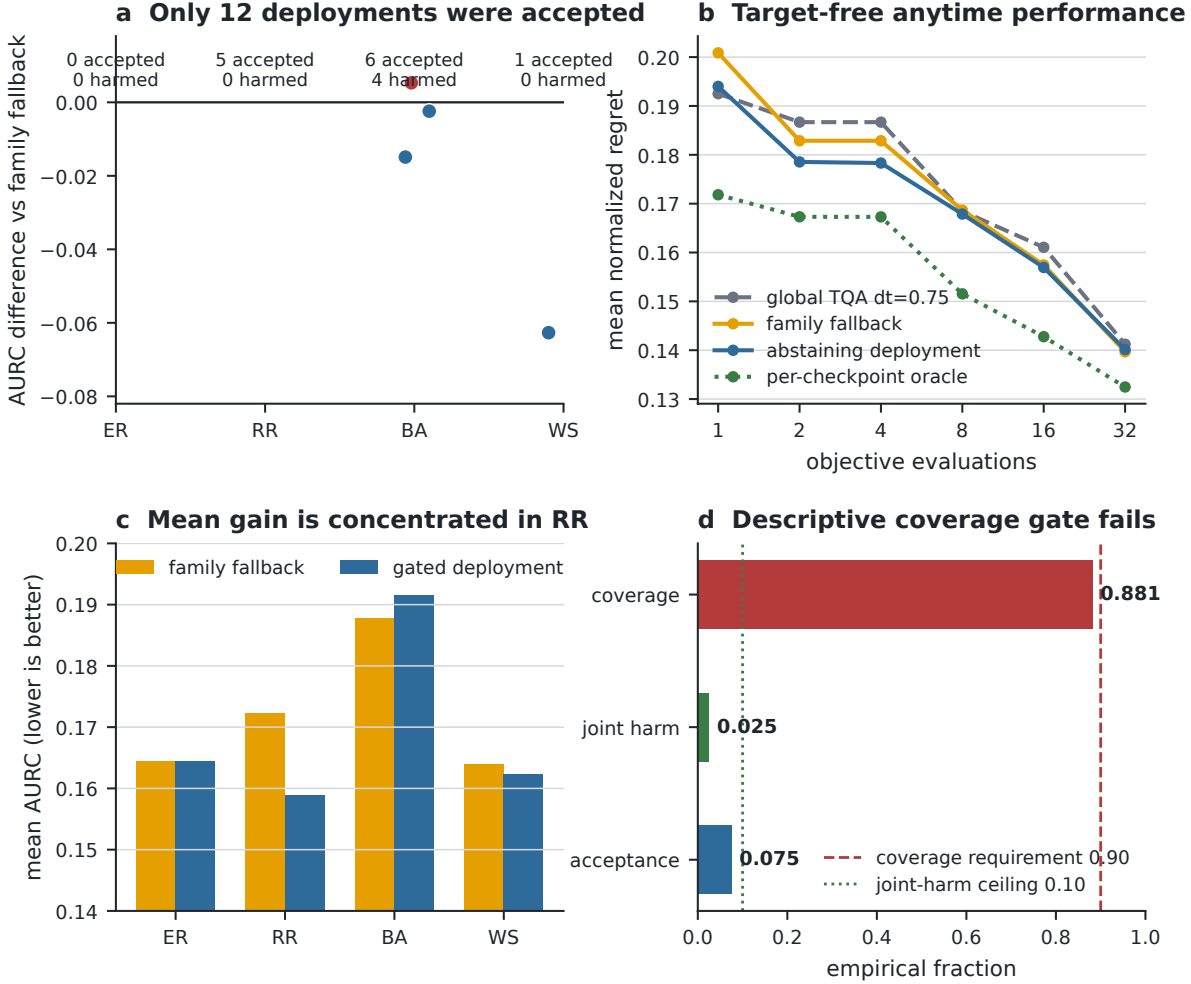


Figure 2: **One-shot confirmatory audit.** The mean gain is small and concentrated by family. Twelve routes are accepted, four are harmful, and descriptive one-sided coverage misses the frozen requirement.

Proposition 4.2 (Harm-rate decomposition). *If $\mathbb{P}(A) > 0$, then*

$$\mathbb{P}(A \cap H) = \mathbb{P}(A)\mathbb{P}(H | A). \quad (10)$$

Consequently, controlling joint harm alone does not control conditional harm when acceptance can be small.

The proof and an explicit counterexample appear in Appendix A.2. In the audit, $\mathbb{P}_n(A) = 12/160 = 0.075$, $\mathbb{P}_n(A \cap H) = 4/160 = 0.025$, and $\mathbb{P}_n(H | A) = 4/12 = 0.333$. The joint rate is small because acceptance is small, not because accepted routes are uniformly safe.

Marginal coverage is also distinct from selective harm. An upper residual bound can cover Δ while the acceptance rule still concentrates errors near the decision boundary. Conversely, a policy may have acceptable joint harm while missing a nominal residual-coverage target. All four quantities must be reported; marginal calibration and conditional selection answer different questions [3, 50, 59].

4.3 Acceptance, Exceedance, and Non-Certification

Write $\hat{d} = \hat{u}_s - \hat{u}_b$, $R = \Delta - \hat{d}$, $A = \{\hat{d} + q < 0\}$, and $H = \{\Delta > 0\}$. The acceptance rule has a useful deterministic implication that does not itself require a sampling model.

Proposition 4.3 (Acceptance–exceedance containment). *For every joint distribution of $(\Delta, \hat{d})$ and every $q \in \mathbb{R}$,*

$$A \cap H \subseteq \{R > q\}. \quad (11)$$

Consequently, for any $\alpha \in [0, 1]$, if $\mathbb{P}(R \leq q) \geq 1 - \alpha$, then $\mathbb{P}(A \cap H) \leq \alpha$; if additionally $\mathbb{P}(A) > 0$, then $\mathbb{P}(H | A) \leq \alpha / \mathbb{P}(A)$.

Appendix A.4 proves every implication. The result clarifies both the appeal and the limitation of residual calibration: idealized marginal residual coverage controls joint harm, whereas the conditional bound deteriorates as

acceptance becomes rare. More importantly, the locked fixed-quota, sequentially filtered audit does not establish the pooled exchangeability needed to infer the premise from the order statistic. Proposition 4.3 is therefore an algebraic interpretation of the gate, not a certificate for the released audit [8].

Dependence induced by fixed family quotas. Balancing an audit across graph families improves the visibility of family-specific failures, but it changes the sampling law. Even if the final order of the audit graphs is randomized, the family labels are dependent once their totals are fixed, so comparisons must respect the data-set and stratum as the sampling units [18, 19].

Proposition 4.4 (Fixed quotas are not pooled iid sampling). *Consider an audit of size n with exactly n_f graphs from family f , where $0 < n_f < n$, and let the order of the n graphs be uniformly randomized. For distinct positions $i \neq j$, define $Z_i = \mathbf{1}\{\text{position } i \text{ has family } f\}$. Then*

$$\text{Cov}(Z_i, Z_j) = -\frac{p_f(1-p_f)}{n-1} < 0, \quad p_f = \frac{n_f}{n}. \quad (12)$$

Consequently, the pooled family labels are exchangeable but not independent, and the audit is not an iid sample from the mixture with family mass p_f .

Appendix A.3 proves the covariance identity by direct counting. The proposition does not invalidate stratum-specific description or a design-based analysis. It limits only an iid interpretation that ignores the fixed quotas and sequential graph filtering.

5 Graph-Conditioned Trust Regions

5.1 Target-Free Portfolio Utility

Every arm receives the same 32-call counter. Seed evaluations, restarts, and all intermediate queries count against the ceiling. The final utility in Eq. (5) rewards useful values early without granting an optimizer access to $C_{\max}$. Exact MaxCut remains a small-instance analysis oracle, so the metric is target free for control but not hardware native.

The nine arms are a training-set concentration vector; four midpoint linear schedules; one-nearest-neighbor transfer; four-start random multistart; SPSSA; and the legacy graph-conditioned Gaussian trust-region policy. A fixed schedule can be competitive because QAOA angles concentrate [2]; the portfolio therefore retains simple arms instead of comparing only learned policies.

5.2 Structural Prediction and Family Fallback

The structural vector contains size, edge count per vertex, density, seven degree summaries, mean clustering,

transitivity, triangles per vertex, and three normalized-Laplacian summaries. A multi-output ridge model predicts each arm’s AUC:

$$\hat{\mathbf{u}}(x) = \mathbf{a} + W\tilde{x}, \quad \lambda_{\text{ridge}} = 1, \quad (13)$$

where normalization uses development means and scales only.

The known synthetic family label defines a strong deployable fallback. This choice is intentionally favorable to the control. The router must improve upon what can be selected without instance-level prediction. General algorithm-selection theory warns that larger portfolios increase both opportunity and overfitting risk [13, 49].

5.3 Residual Order-Statistic Abstention

Fix $\alpha \in (0, 1)$. Suppose calibration residuals $R_1, \dots, R_n$ and a future residual R_{n+1} are exchangeable and almost surely distinct. Let $k = \lceil (n+1)(1-\alpha) \rceil$, and let q be the k th calibration order statistic, with $q = +\infty$ if $k > n$.

Theorem 5.1 (Split order-statistic coverage). *Under the preceding assumptions,*

$$\mathbb{P}(R_{n+1} \leq q) = \frac{k}{n+1} \geq 1 - \alpha. \quad (14)$$

Appendix A.5 proves the result by rank symmetry. Ties can be handled conservatively or by randomized ranks. The theorem explains the calibration rule; it does not assert that the quota-filtered audit residual is exchangeable with pooled calibration residuals.

5.4 Locked Confirmatory Decision

The configuration, graph identities, angle labels, implementation, non-audit traces, learned state, calibration statistic, and decision are hashed before audit evaluation. The evidence directory rejects a second audit run. The generator validates these hashes before producing manuscript displays.

This locking scheme is not an external preregistration and does not prevent all researcher degrees of freedom. The hashes establish internal consistency with the declared execution order, and the output lock refuses an overwrite inside the designated evidence directory. They are not an independent timestamp and cannot exclude unrecorded alternative runs. Reproducibility concerns both computational determinism and inferential chronology.

6 Experiments

6.1 Development Selection

Graphs come from four synthetic families: Erdős–Rényi [21], random regular as generated by NetworkX [24], Barabási–Albert [7], and Watts–Strogatz [60]. Sequential filtering rejects isomorphic repeats within each stratum.

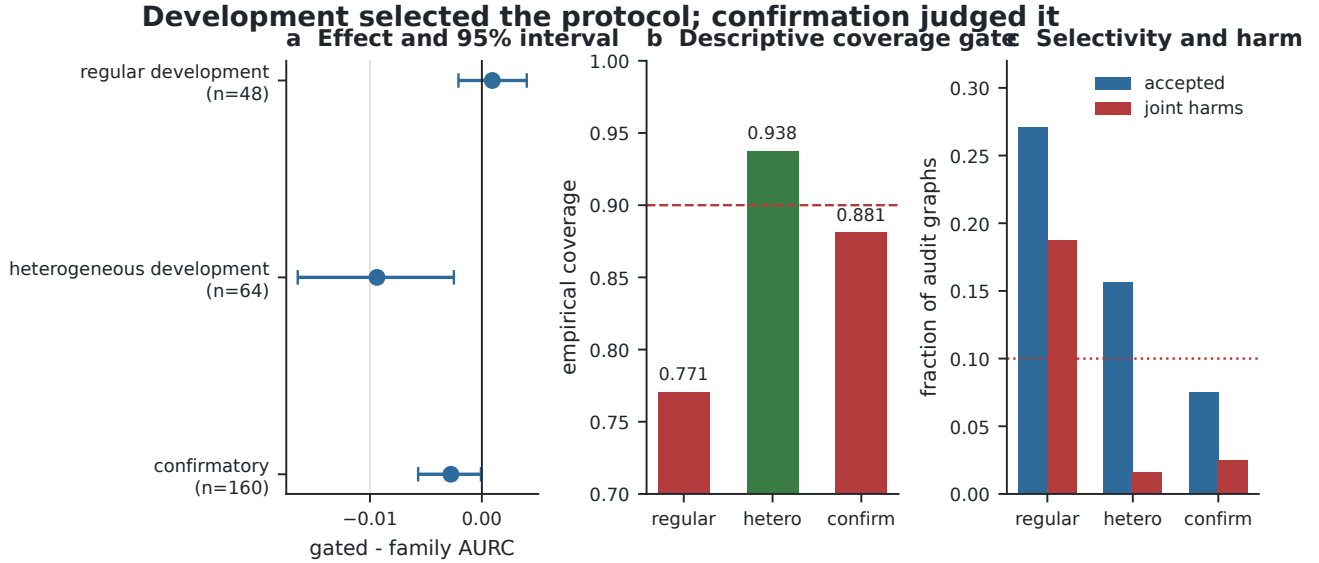


Figure 3: **Development-to-confirmation progression.** The heterogeneous development split satisfies the frozen gate, while the confirmatory split misses its empirical-coverage component. Audit outcomes were not used to retune the released decision.

Separate angle-label, development, calibration, and audit splits are frozen by graph identity.

The initial fixed-parameter generator-grid study included all four families but did not vary each family’s generator parameters broadly. A subsequent 64-graph heterogeneous development-stage audit used declared parameter ranges and triggered the family-fallback protocol. The locked selector was then fit on 48 development graphs and its residual threshold was computed from 48 disjoint calibration graphs. This adaptation occurred before the confirmatory graph list and decision were frozen.

For any split of N graphs, let $\bar{u}_{\text{global}}$ be the mean AURC of the single global arm and let $\bar{u}_{\text{oracle}} = N^{-1} \sum_{i=1}^N \min_a u_{ia}$ be the mean per-graph portfolio oracle. We define portfolio opportunity as

$$\mathcal{O} = \frac{\bar{u}_{\text{global}} - \bar{u}_{\text{oracle}}}{\bar{u}_{\text{global}}}. \quad (15)$$

The heterogeneous development-stage audit recorded $\mathcal{O} = 0.10710$. The locked confirmatory audit retained the requirement $\mathcal{O} \geq 0.05$ and recorded $\mathcal{O} = 0.07115$.

6.2 Confirmatory Audit

The one-shot audit contains 160 graphs, 40 per family. The gated deployment has mean AURC 0.16930, compared with 0.17207 for the family fallback and 0.17277 for the global development arm. The selected-minus-family difference is -0.002768 . A frozen 4,000-resample stratified percentile bootstrap interval is $[-0.005696, -0.00008178]$.

That interval is not the decision by itself. The policy accepts 12 routes, incurs 4 joint harms, and has empirical residual coverage 0.88125 against nominal 0.90. Since

coverage is below the internally frozen threshold, the reliability gate fails. A binomial sensitivity model gives probability 0.249 of observing at most 141 covered cases under independent 0.90 coverage, so the miss is compatible with nominal variation; it nevertheless fails the literal frozen rule.

6.3 Anytime Behavior and Resource Accounting

Every arm receives 32 exact objective calls, so the audit consumes 288 calls per graph across nine arms and 46,080 calls for the 160 audit graphs. The angle-label track is reported separately. Runtime is descriptive because Python and simulator overhead differ by arm.

The released simulator uses PyTorch for numerical kernels [45]. It does not include shot noise, gate noise, compilation, or queue latency. Analytic gradients and higher derivatives can be estimated on quantum hardware under appropriate gates [39], and quantum-natural-gradient methods alter the local metric [56]; neither is evaluated here.

The audit’s statistical unit is the graph. Bootstrap resampling is stratified by the eight family-by-size strata [19]. The fixed quota improves balance but means the pooled sample is not an iid draw from an unspecified graph population. The reported interval is therefore the output of the frozen analysis, not a universal confidence statement.

Table 1: **Frozen data roles and exact-query counts.** Each role uses a disjoint graph split.

Split	Per family-size	Graphs	Exact queries	Role
Angle labels	6	48	5,760	legacy-arm training
Development	6	48	13,824	routing and fallback
Calibration	6	48	13,824	residual order statistic
Confirmatory audit	20	160	46,080	one-shot decision

Arm ranking is stable on average but heterogeneous by graph

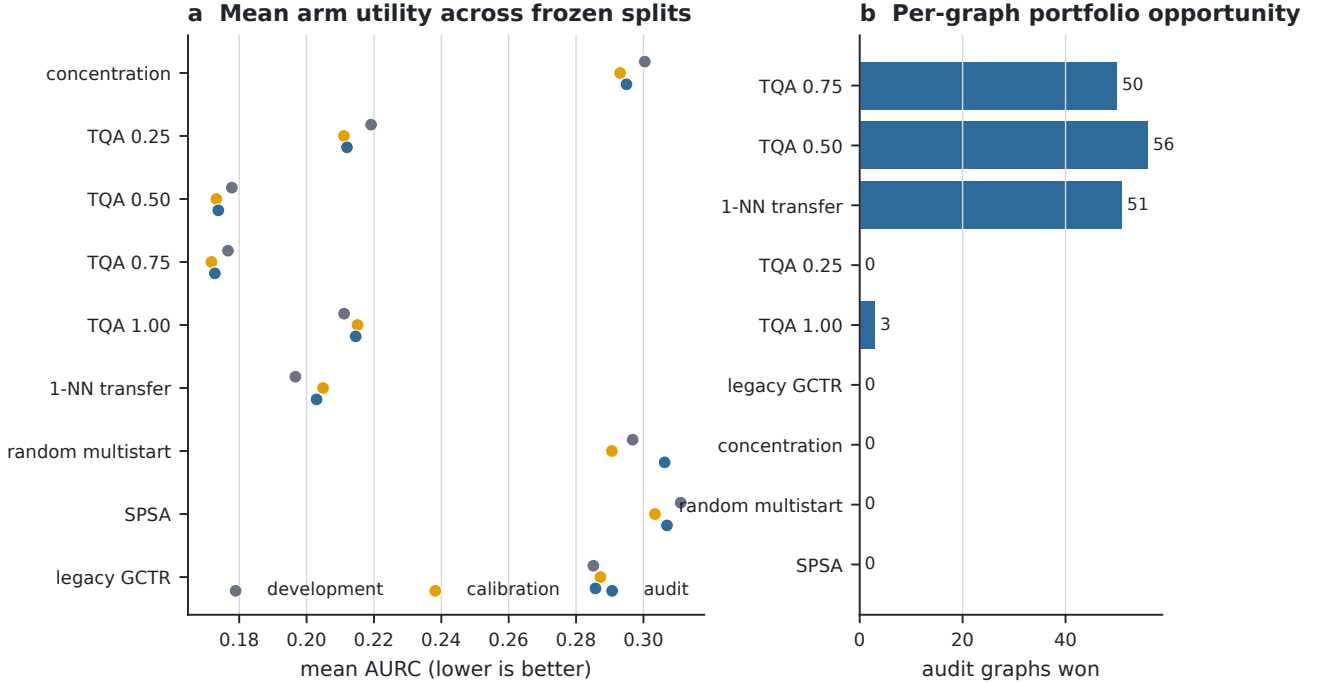


Figure 4: **Arm opportunity and split stability.** (a) Mean AURC for every arm on the frozen development, calibration, and audit splits; lower is better. (b) Number of audit graphs on which each arm has the smallest AURC. Heterogeneous graph-level winners motivate a portfolio, but do not by themselves validate a learned router.

7 Conclusion

Graph structure contains useful information about QAOA optimizer behavior, but useful information is not the same as reliable routing. The frozen selector slightly improves mean anytime regret over a strong family fallback, yet its accepted set is small, one third of accepted routes are harmful, and the internally frozen empirical-coverage requirement fails. We therefore reject the confirmatory reliability claim.

Limitations. The experiment has six major limitations. It uses exact statevectors, depth two, small unweighted MaxCut graphs, fixed graph-family quotas, a known family label at deployment, and a hand-designed nine-arm portfolio. It provides no hardware result, noise result, asymptotic scaling result, or quantum advantage. The percentile bootstrap endpoint is boundary sensitive,

and the pooled exchangeability condition required by the simplest order-statistic theorem is not established.

The next experiment should replace the descriptive gate with a prospective sampling design, predefine the graph population and filtering procedure, increase accepted-route power, include shot and device noise, and compare against stronger adaptive controls. If multiple reliability criteria are tested, the associated multiplicity correction and joint decision rule should be fixed before analysis [9]. If exact binomial or concentration bounds are reported, the associated independence or exchangeability assumptions must be stated [15, 27].

The repository is an installable package. The command `gctr-reproduce -validate-only` audits source records and hashes; `gctr-reproduce` regenerates the original data displays; and `python manuscript/generate_portfolio_artifacts.py` re-

Table 2: **Confirmatory results by graph family.** Lower AURC is better and Δ is gated minus family fallback.

Stratum	N	Fallback	Gated	Δ	Accepted	Harms
Erdős–Rényi	40	0.16439	0.16439	+0.00000	0	0
random regular	40	0.17223	0.15896	-0.01327	5	0
Barabási–Albert	40	0.18775	0.19151	+0.00376	6	4
Watts–Strogatz	40	0.16391	0.16234	-0.00157	1	0
All	160	0.17207	0.16930	-0.00277	12	4

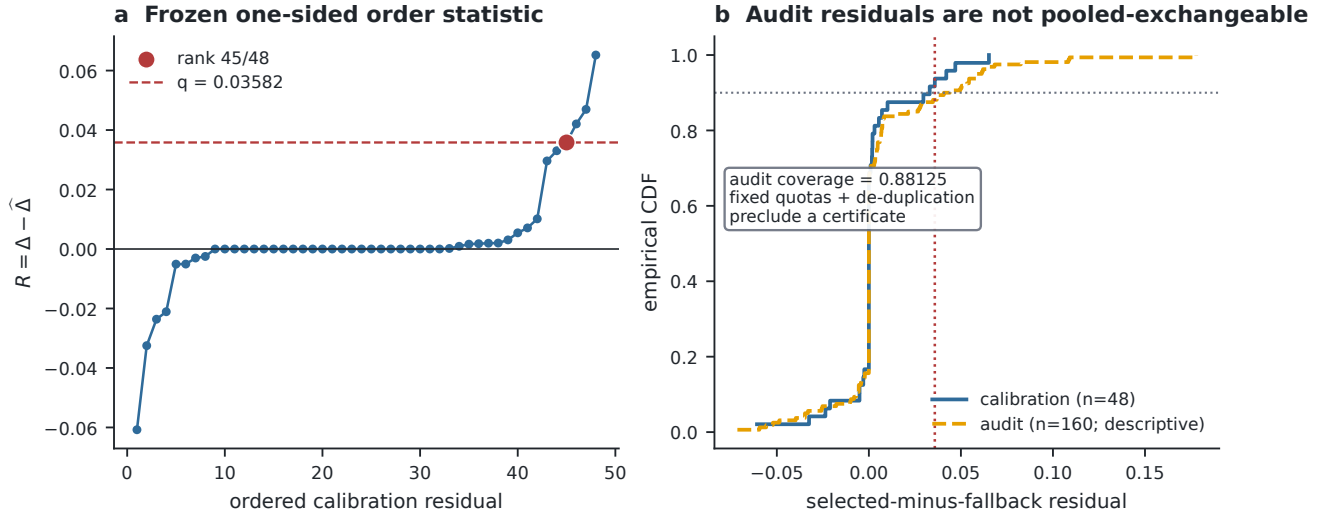
Calibration defines a frozen heuristic, not a realized coverage theorem

Figure 5: **Frozen residual rule and its inferential boundary.** (a) The one-sided calibration statistic is the rank-45 order statistic among 48 residuals. (b) Calibration and audit empirical CDFs are shown descriptively. Fixed family quotas and sequential de-duplication do not justify a pooled exchangeability certificate.

generates the locked portfolio figures and tables. A separate prospective-risk-control design file is marked unregistered, incomplete, and unexecuted; it contributes no result.

Predicted improvements do not uniformly transfer to accepted graphs

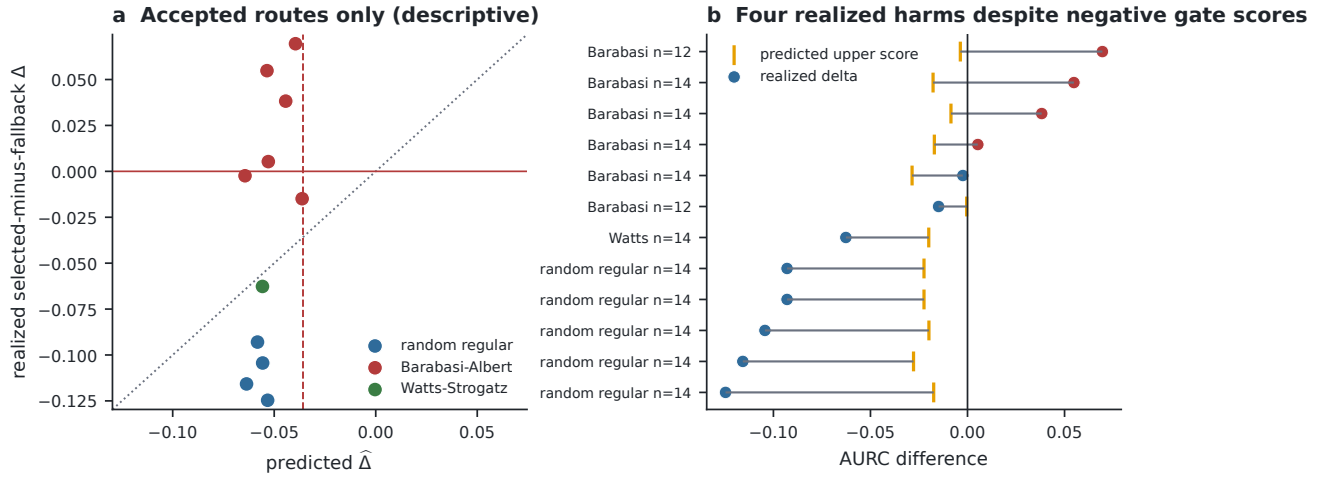


Figure 6: **Predicted versus realized accepted routes.** (a) Predicted and realized selected-minus-fallback AURC differences for the 12 accepted graphs. (b) Four accepted routes are harmful despite negative frozen upper scores. The display diagnoses selective failure; it does not refit the gate.

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A Proofs

A.1 Proof for Theorem 4.1

Proof. Let the instance space contain two points x_0 and x_1 . Define losses

$$\begin{aligned}\ell(a, x_0) &= 0, & \ell(b, x_0) &= \varepsilon, \\ \ell(a, x_1) &= 1, & \ell(b, x_1) &= \varepsilon.\end{aligned}$$

Every displayed loss belongs to $[0, 1]$ because $\varepsilon \in (0, 1)$. Let P_0 put probability one on x_0 and P_1 put probability one on x_1 . Under P_0 ,

$$\mathbb{E}_{P_0} \ell(a, X) = 0 < \varepsilon = \mathbb{E}_{P_0} \ell(b, X),$$

so a is the unique development risk minimizer. Under P_1 ,

$$\mathbb{E}_{P_1} \ell(b, X) = \varepsilon < 1 = \mathbb{E}_{P_1} \ell(a, X),$$

so b is the unique deployment risk minimizer. A router that unconditionally deploys the development minimizer a has deployment excess risk

$$\mathbb{E}_{P_1} \ell(a, X) - \min_{c \in \{a, b\}} \mathbb{E}_{P_1} \ell(c, X) = 1 - \varepsilon.$$

Because ε can approach zero, this excess approaches the largest possible difference for losses in $[0, 1]$. No estimation error or model misspecification is needed. $\square$

A.2 Proof for Proposition 4.2

Proof. By the definition of conditional probability, when $\mathbb{P}(A) > 0$,

$$\mathbb{P}(H \mid A) = \frac{\mathbb{P}(A \cap H)}{\mathbb{P}(A)}.$$

Multiplying both sides by $\mathbb{P}(A)$ gives Eq. (10).

To prove the second claim, fix any joint-harm ceiling $\eta \in (0, 1)$ and any $c \in [\eta, 1]$. Construct events with $\mathbb{P}(A) = \eta/c$ and $\mathbb{P}(H \mid A) = c$. Then $\mathbb{P}(A \cap H) = \eta$ while the conditional harm is c . In particular, taking $c = 1$ and $\mathbb{P}(A) = \eta$ gives joint harm η with every accepted route harmful. Thus a bound on $\mathbb{P}(A \cap H)$ alone cannot imply a nontrivial bound on $\mathbb{P}(H \mid A)$ unless acceptance is bounded away from zero. $\square$

A.3 Proof for Proposition 4.4

Proof. Uniformly randomizing the audit order is equivalent to choosing the n_f positions occupied by family f uniformly from the $\binom{n}{n_f}$ subsets of size n_f . For any position i ,

$$\mathbb{P}(Z_i = 1) = \frac{\binom{n-1}{n_f-1}}{\binom{n}{n_f}} = \frac{n_f}{n} = p_f.$$

For distinct positions i and j ,

$$\mathbb{P}(Z_i = 1, Z_j = 1) = \frac{\binom{n-2}{n_f-2}}{\binom{n}{n_f}} = \frac{n_f(n_f-1)}{n(n-1)}.$$

Here and below, $\binom{N}{k} = 0$ when $k < 0$ or $k > N$; thus the displayed identity also covers $n_f = 1$. Therefore

$$\begin{aligned}\text{Cov}(Z_i, Z_j) &= \mathbb{E}[Z_i Z_j] - \mathbb{E}[Z_i] \mathbb{E}[Z_j] \\ &= \frac{n_f(n_f-1)}{n(n-1)} - \frac{n_f^2}{n^2} \\ &= -\frac{n_f(n-n_f)}{n^2(n-1)} = -\frac{p_f(1-p_f)}{n-1}.\end{aligned}$$

Because $0 < n_f < n$, the covariance is strictly negative. Independent Bernoulli variables have zero covariance, so the family indicators cannot be independent. Their joint law is invariant under permutations of audit positions, hence exchangeable. Finally, iid sampling from a mixture with family probability p_f would produce a random family count with binomial variance $np_f(1-p_f) > 0$, whereas the quota design fixes that count at n_f with variance zero. Thus the quota design is not pooled iid mixture sampling. $\square$

A.4 Proof for Proposition 4.3

Proof. Take any outcome in $A \cap H$. Membership in A gives

$$\hat{d} + q < 0,$$

and hence

$$-\hat{d} > q.$$

Membership in H gives $\Delta > 0$. Therefore

$$R = \Delta - \hat{d} > -\hat{d} > q.$$

Thus every outcome in $A \cap H$ belongs to $\{R > q\}$, which proves Eq. (11).

Taking probabilities on both sides of the set inclusion yields

$$\mathbb{P}(A \cap H) \leq \mathbb{P}(R > q) = 1 - \mathbb{P}(R \leq q).$$

If $\mathbb{P}(R \leq q) \geq 1 - \alpha$, then the right-hand side is at most α , so $\mathbb{P}(A \cap H) \leq \alpha$. Finally, when $\mathbb{P}(A) > 0$,

$$\mathbb{P}(H \mid A) = \frac{\mathbb{P}(A \cap H)}{\mathbb{P}(A)} \leq \frac{\alpha}{\mathbb{P}(A)}.$$

Each step follows from the deterministic containment and the definition of conditional probability; no exchangeability assumption is used inside this proof. Exchangeability is relevant only to establishing the separate premise $\mathbb{P}(R \leq q) \geq 1 - \alpha$ from calibration data. $\square$

A.5 Proof for Theorem 5.1

Proof. Because $R_1, \dots, R_{n+1}$ are exchangeable and almost surely distinct, the rank

$$\text{rank}(R_{n+1}) = 1 + \sum_{i=1}^n \mathbf{1}\{R_i < R_{n+1}\}$$

is uniform on $\{1, \dots, n+1\}$. Let q be the k th smallest calibration score. If $k = n+1$, the convention $q = +\infty$ gives $\mathbb{P}(R_{n+1} \leq q) = 1 = k/(n+1)$, and the claim follows. Hence assume $1 \leq k \leq n$. If the test rank is at most k , then at most $k-1$ calibration scores are smaller than R_{n+1} , so the k th calibration score is at least R_{n+1} and $R_{n+1} \leq q$. If the test rank exceeds k , then at least k calibration scores are smaller, so $R_{n+1} > q$. Hence

$$\{R_{n+1} \leq q\} = \{\text{rank}(R_{n+1}) \leq k\}.$$

Uniformity of the rank gives

$$\mathbb{P}(R_{n+1} \leq q) = \frac{k}{n+1}.$$

Finally, $k = \lceil (n+1)(1-\alpha) \rceil \geq (n+1)(1-\alpha)$, so division by $n+1$ gives the desired lower bound. $\square$

B Algorithm

Algorithm 1: Frozen structural routing

Given a graph G , development standardization, ridge coefficients, a family fallback map, and order statistic q :

1. compute the 16 relabeling-invariant summaries $x(G)$;
 2. standardize them using development means and scales;
 3. predict all nine AURCs with Eq. (13);
 4. choose $s = \arg \min_a \hat{u}_a(x)$ and retrieve $b(f)$;
 5. accept s only if $\hat{u}_s - \hat{u}_b + q < 0$; otherwise deploy b ;
 6. run the deployed arm with a hard 32-call counter and record every best-so-far checkpoint.
-

The audit implementation additionally verifies that the graph, configuration, decision, and implementation hashes match the frozen manifest before evaluation.

Table 3: **Frozen nine-arm portfolio.** Every arm receives the same 32-call ceiling; offline labels and search rules are shown explicitly.

Arm	Initialization	Search rule	Offline labels	Calls
Concentration	coordinate-wise median label	pattern search	yes	32
TQA $d_t = 0.25$	midpoint linear ramp	pattern search	no	32
TQA $d_t = 0.50$	midpoint linear ramp	pattern search	no	32
TQA $d_t = 0.75$	midpoint linear ramp	pattern search	no	32
TQA $d_t = 1.00$	midpoint linear ramp	pattern search	no	32
1-NN transfer	nearest standardized training graph	pattern search	yes	32
Random multistart	four uniform random starts	four pattern searches	no	32
SPSA	concentration angle	$a = 0.18, c = 0.12, \alpha = 0.602, \gamma = 0.101$	yes	32
Legacy GCTR	GIN diagonal-Gaussian mean	radius-two covariance-shaped pattern search	yes	32

C Optimizer Portfolio

The concentration arm uses the coordinate-wise median of the 48 angle-label training vectors. Four linear schedules interpolate fixed midpoint ramps. One-nearest-neighbor transfer retrieves the closest angle-label training graph in standardized feature space. Multistart divides the budget across four seeded starts. SPSA uses a frozen gain schedule. The legacy GCTR arm maps graph features to a Gaussian angle policy and a local budget rule.

The portfolio mixes model-based and model-free search. Gaussian-process optimization is a natural future arm [6, 54]; randomized low-rank models may reduce structural fitting cost on much larger feature sets [40]. Neither is needed for the released 16-feature ridge fit.

D Bias and Variance of Routing Estimates

Ridge predictions have estimation error from finite development data, model bias from linearity, and distribution shift across graph strata. The selected-minus-fallback residual further contains selection-induced dependence because the selected arm is an argmin of the same prediction vector. Calibration treats the resulting scalar residual as the conformity score without assuming Gaussianity.

The frozen bootstrap resamples graphs within each family-by-size stratum. Its percentile endpoint is sensitive when the observed upper bound is close to zero. Normal, studentized, or randomization-based alternatives would answer different questions and were not substituted after seeing the audit.

E Graph-Conditioned Optimizer Arms

E.1 Concentration Schedule

The coordinate-wise median of the 48 angle-label training vectors is a zero-query structural control after training.

E.2 Linear Ramps

Four fixed midpoint schedules span conservative to aggressive angle scales.

E.3 Nearest-Neighbor Transfer

The nearest standardized angle-label training feature vector supplies an angle seed.

E.4 Random Multistart

Four independent starts divide the same 32-call budget.

E.5 SPSA

Two perturbed objectives estimate a stochastic search direction at each update.

Anytime behavior differs across the four fixed graph families

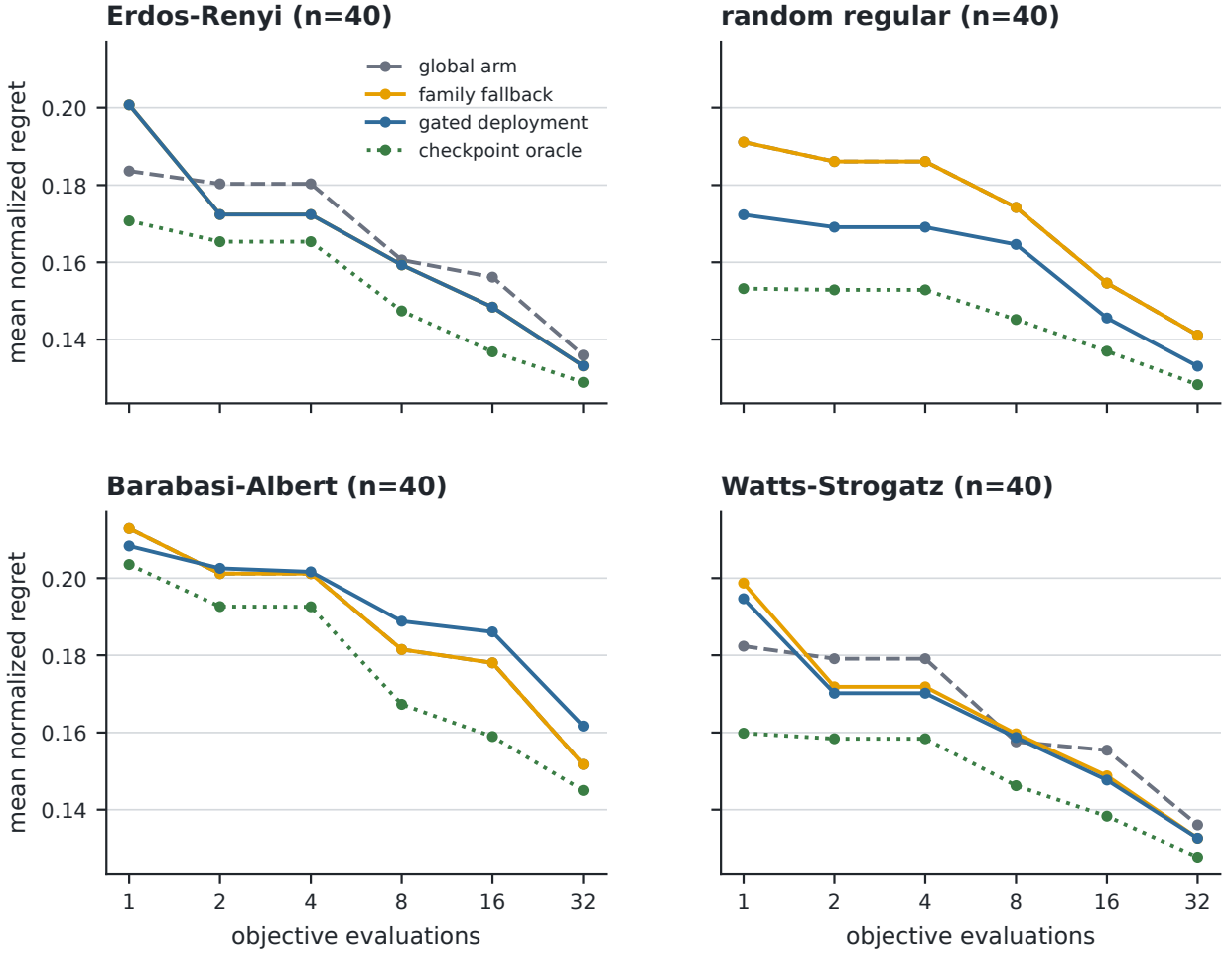


Figure 7: **Anytime behavior by fixed graph family.** Mean normalized regret is shown at every recorded query checkpoint for the global arm, family fallback, gated deployment, and per-checkpoint oracle. Family heterogeneity persists throughout the 32-call budget.

E.6 Legacy Trust Region

The graph-conditioned Gaussian supplies an initializer, covariance-shaped region, and frozen local policy. It receives no extra calls.

E.7 Family Fallback

The lowest development mean-AURC arm within the known generator family is the deployable control.

E.8 Gated Selector

The structural argmin replaces the fallback only when Eq. (9) certifies a negative predicted upper bound under the frozen score.

F Graph Benchmark Tasks

F.1 Erdős–Rényi

Edges are sampled independently before sequential non-isomorphism filtering.

Pooled improvement masks concentrated benefit and harm

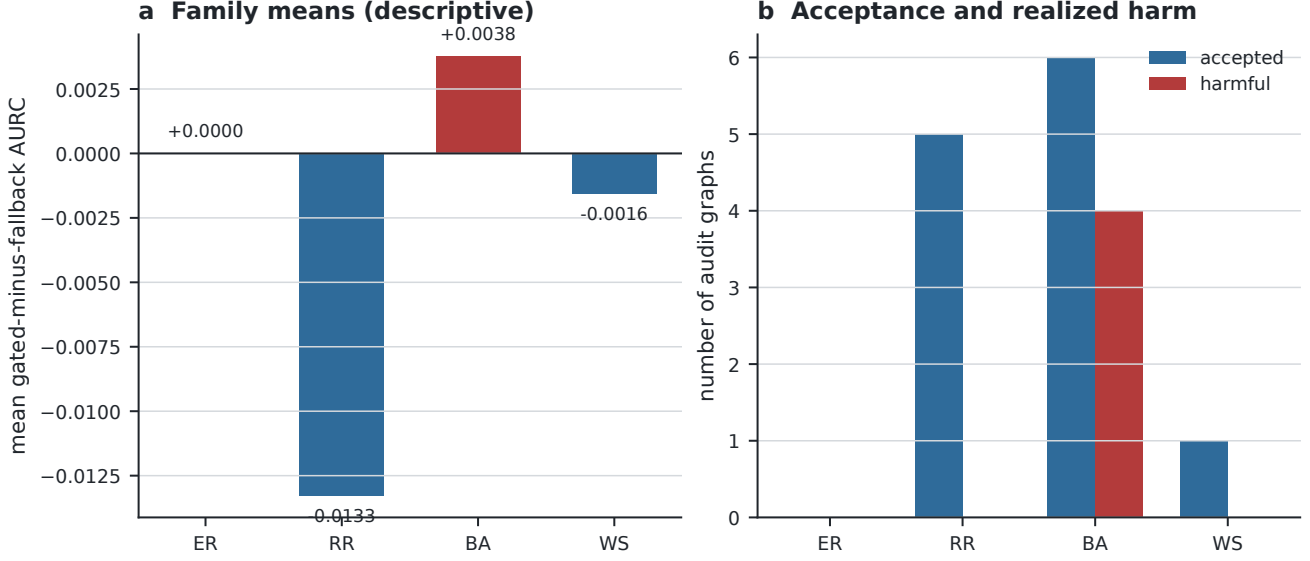


Figure 8: **Family-conditional effects.** (a) Mean gated-minus-fallback AURC within each fixed family. (b) Acceptance and realized-harm counts. The pooled improvement is concentrated in random regular graphs, whereas all four observed harms occur in the Barabási–Albert stratum.

F.2 Random Regular

Every vertex has fixed degree, producing a homogeneous structural stratum.

F.3 Barabási–Albert

Preferential attachment yields heterogeneous degrees and hub structure.

F.4 Watts–Strogatz

Ring-lattice rewiring yields clustered small-world graphs.

F.5 Fixed-Quota Audit

Each family contributes exactly 40 audit graphs. The balanced design supports stratified description but is not a pooled iid sample from a naturally occurring graph population.

G Experimental Details for Development

Development and calibration each use disjoint graph lists. Feature standardization, ridge fitting, family-arm means, and residual sorting use only their assigned roles. The calibration quantile is the conservative upper order statistic for $\alpha = 0.10$.

The gate table is kept distinct from interval estimation: it applies the recorded decision rule literally to the locked summaries and does not reinterpret a failed criterion after audit inspection.

H Experimental Details for the Locked Audit

The confirmatory configuration fixes 20 graphs per family-size stratum for 160 total. The decision manifest stores the selected arm, fallback, predicted delta, acceptance flag, and relevant hashes before audit traces are scored. Non-audit traces remain available for chronology checks.

Observed rates and confidence bounds answer different questions

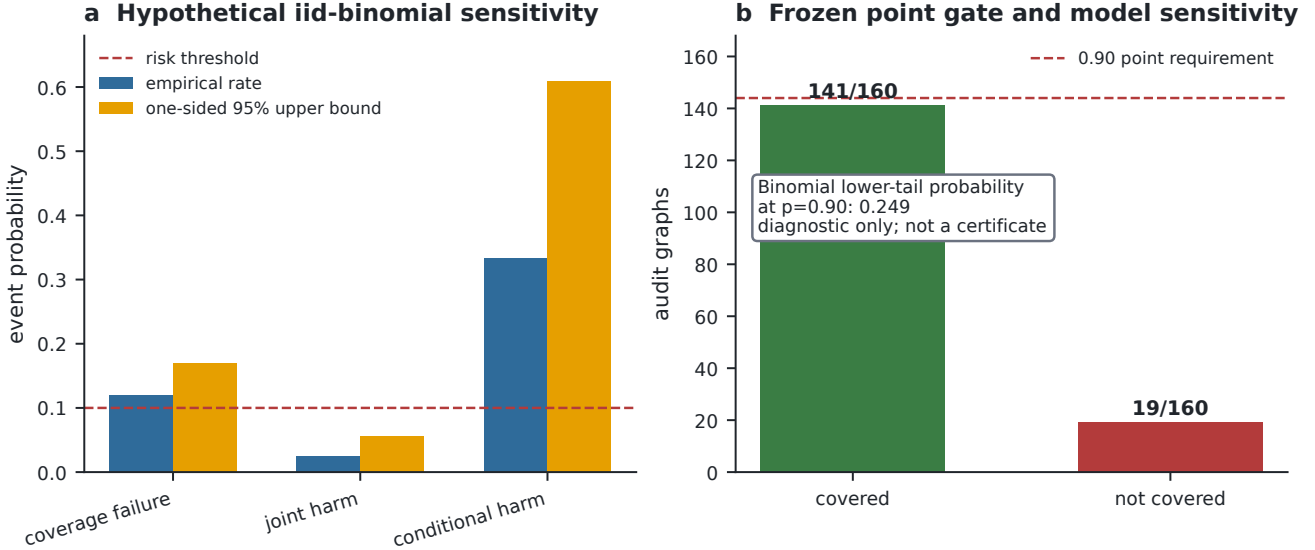


Figure 9: **Risk quantities are not interchangeable.** (a) Empirical coverage-failure, joint-harm, and conditional-harm rates are paired with one-sided exact-binomial upper bounds under a hypothetical iid model. (b) The observed coverage count is compared with the frozen point requirement. These calculations are sensitivity analyses, not certificates for the dependent fixed-quota design.

I Resource Accounting

Every optimizer arm receives exactly 32 objective calls. Fixed schedules still pay for their seed evaluation. Multistart pays for each start from the same counter. Model training and angle-label generation are reported outside the deployable per-graph budget. Exact statevector runtime is not converted into hardware shots.

The displayed ledger covers the final locked protocol and totals 79,488 exact objective calls. The earlier regular-grid and heterogeneous development-stage studies used 36,096 and 51,840 additional calls, respectively. Repeating all three committed evidence tracks would therefore use 167,424 exact objective calls. These are research-construction costs, not the 32-call cost of deploying one frozen arm on a new graph.

J Additional Reliability Analysis

The 141 covered residuals out of 160 fall below the literal 0.90 requirement. Under a hypothetical independent binomial model with success probability 0.90, the lower-tail probability is 0.249. This sensitivity analysis explains that the miss is not surprising; it does not alter the frozen pass/fail rule or justify independence.

The legacy optimizer-ablation values remain reproducible from `tables/table_ablation.tex`. They are historical development diagnostics rather than part of the locked confirmatory portfolio decision, so they are excluded from the five numbered study tables.

Future reliability work should power the accepted subset directly, report family-conditional coverage, and predefine how graph filtering affects the sampling law. Shadow-tomography and classical-shadow measurement methods could support broader operator portfolios [1, 29], but the present commuting MaxCut objective does not require them.

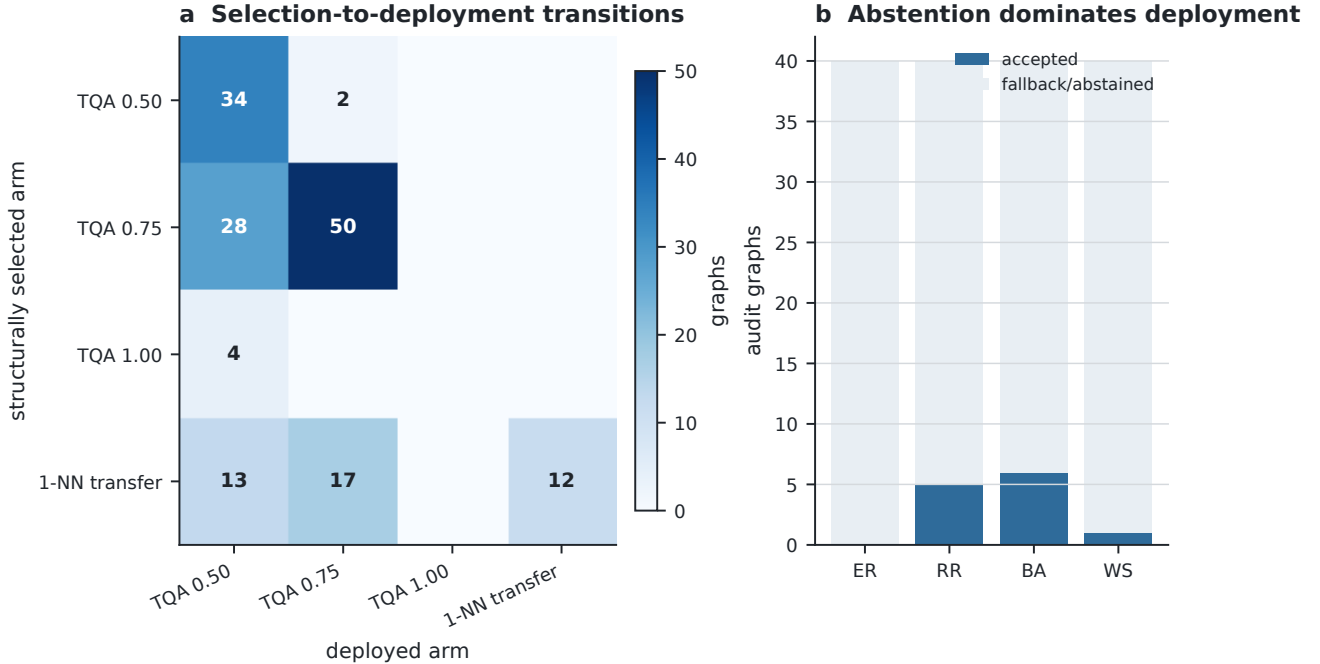
Table 4: **Frozen confirmatory gate.** The empirical-coverage component fails, so the overall decision is no-go.

Criterion	Required	Observed	Result
Portfolio opportunity	≥ 0.05	0.071	pass
Global-baseline interval upper end	< 0	-0.000646	pass
Family-fallback interval upper end	< 0	-0.00008178	pass
Accepted audit graphs	≥ 10	12	pass
One-sided empirical coverage	≥ 0.90	0.88125	fail
Joint harm rate	≤ 0.10	0.0250	pass

Table 5: **Final-protocol exact-statevector resource ledger.** Query totals are regenerated from the locked confirmatory summary. Earlier development-stage studies are reported in the surrounding text, and the deployed per-graph ceiling is not multiplied by the nine-arm research audit.

Stage	Graphs	Exact objective calls	Accounting role
Angle-label training	48	5,760	legacy-arm labels
Development traces	48	13,824	nine-arm portfolio
Calibration traces	48	13,824	nine-arm portfolio
Confirmatory traces	160	46,080	nine-arm audit
Deployed policy	1	32	per new graph

The gate converts most structural proposals back to family controls

Figure 10: **Selection-to-deployment transitions.** (a) Counts from the structurally selected arm to the actually deployed arm. (b) Accepted and abstained graphs by family. The gate sends most structural proposals back to their family control.

Balanced strata aid diagnosis but do not imply pooled iid sampling

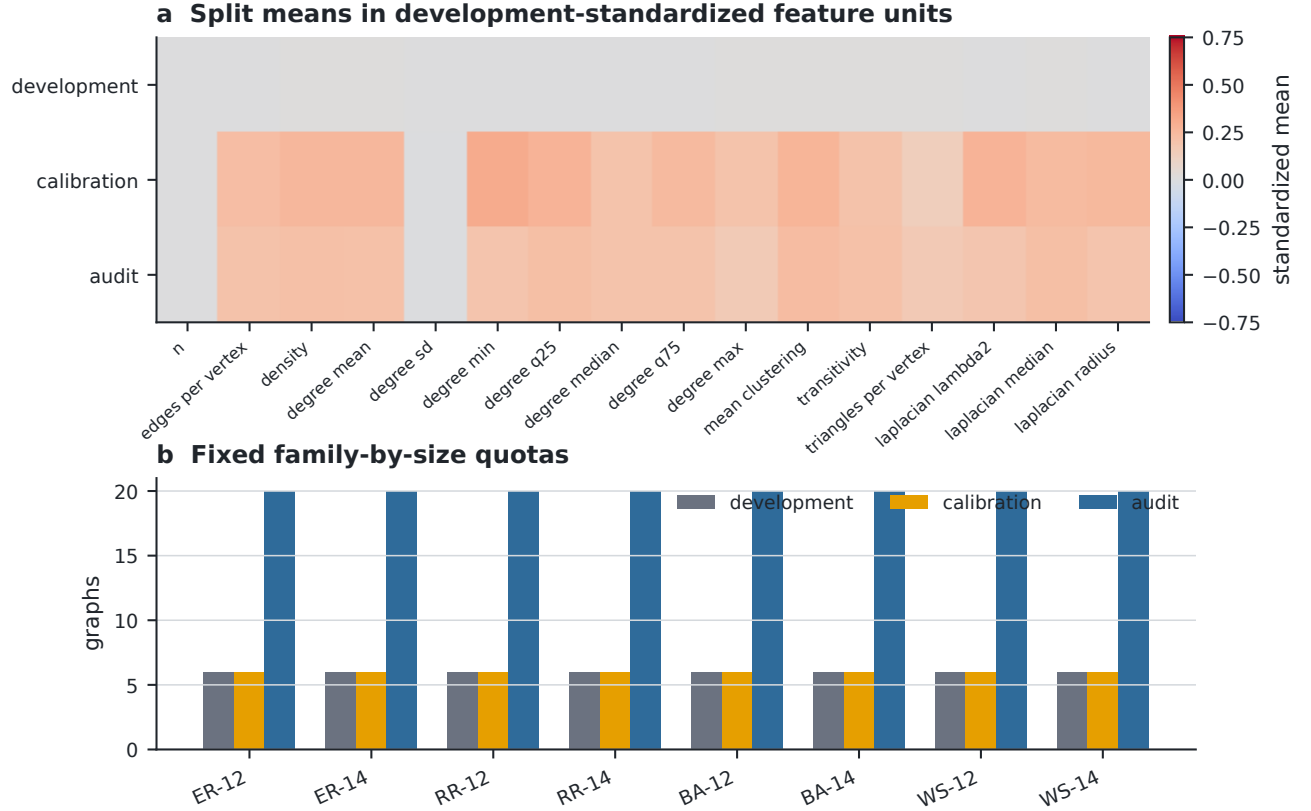


Figure 11: **Split diagnostics.** (a) Feature means expressed in development-standardized units for development, calibration, and audit. (b) Fixed family-by-size quotas. Balance aids stratified diagnosis but does not make the pooled sample iid.

Equal online budgets do not erase offline experiment-construction cost

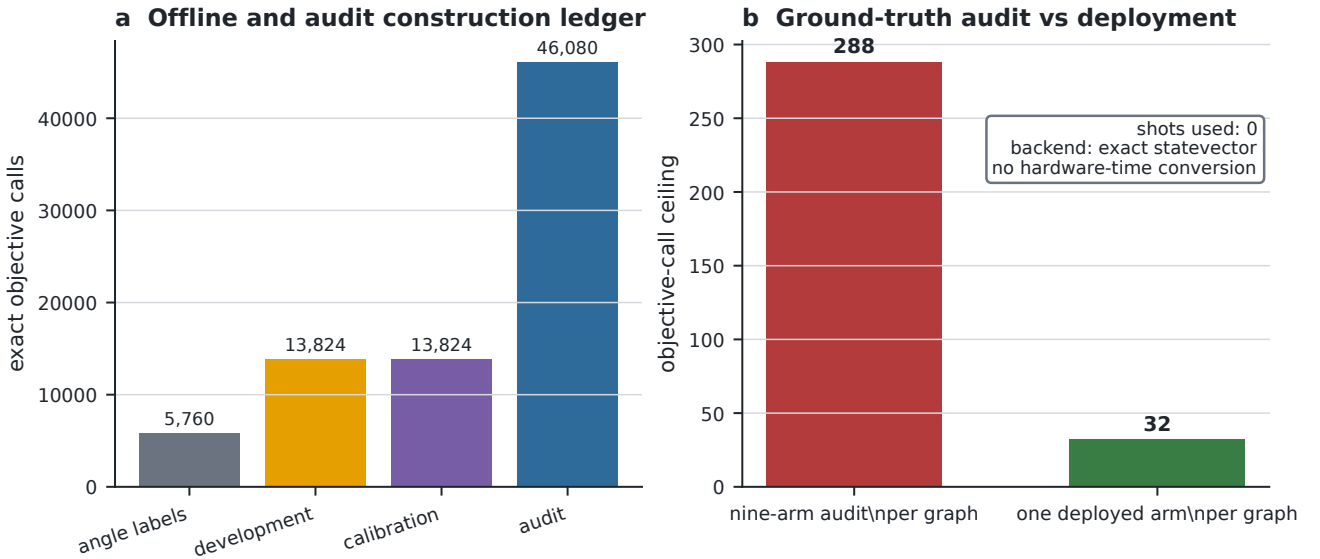


Figure 12: **Exact-statevector resource ledger.** (a) Exact objective calls used for angle labels, development, calibration, and the one-shot audit. (b) The nine-arm research audit is separated from the 32-call ceiling of one deployed arm. No conversion to shots or hardware time is made.

Sensitivity diagnostics do not replace the frozen confirmatory decision

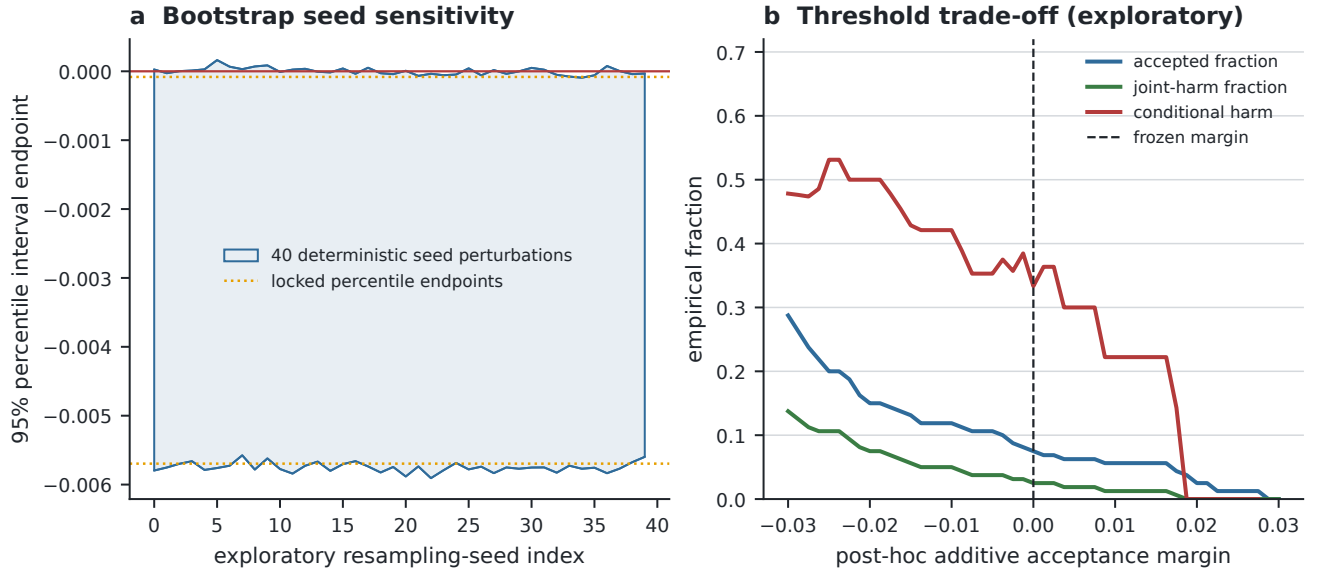


Figure 13: **Exploratory inference and threshold sensitivity.** (a) Percentile-bootstrap endpoints under 40 deterministic resampling seed perturbations. (b) Acceptance and harm rates under post-hoc additive gate margins. Neither panel replaces or modifies the frozen confirmatory decision.